BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY

Department of Electrical and Electronic Engineering

Course No : EEE 212
Course Title: Numerical Technique Laboratory

Project Report

Project Name: Animated Double Pendulum

Submitted by:

Name: Md. Nagib Mahfuz
ID: 2006166

Name: Md. Hamidur Rahman
ID: 2006167

Department: EEE
Section: C2
Group: 2
Level: 2
Term: 1

Date of Submission: 09/03/2023

Submitted to:

1. Bejoy Sikder
Lecturer
Department of Electrical and Electronic Engineering
Bangladesh University of Engineering and Technology

2. Meftahul Haque
Teaching Assistant
Department of Electrical and Electronic Engineering
Bangladesh University of Engineering and Technology

Introduction:

A double pendulum is a simple physical system consisting of two masses connected by rigid rods that are free to swing back and forth. Despite its apparent simplicity, the double pendulum exhibits complex and chaotic behavior, making it a popular topic in physics and mathematics.

In a double pendulum project in MATLAB, we can simulate the motion of a double pendulum using numerical methods to solve the differential equations that govern its motion. The project involves implementing the equations of motion, choosing appropriate initial conditions, and using MATLAB's numerical methods to simulate and visualize the motion of the system.

One way to approach the project is to use the Lagrangian formulation of mechanics to derive the equations of motion for the double pendulum. You can then use Euler’s Method and Improved Euler’s Method to numerically solve the differential equations and simulate the motion of the system. The simulation can be visualized using MATLAB's graphics tools, such as the plot function, to create animations of the double pendulum's motion.

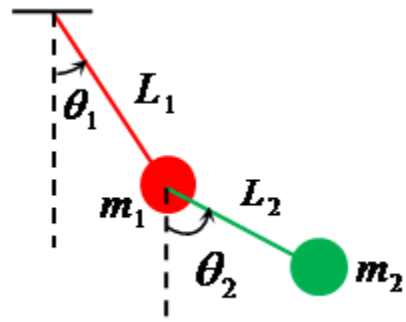
In addition to simulating the motion of the double pendulum, you can also explore the system's behavior under different initial conditions, such as different starting angles or initial velocities. This can help you gain a better understanding of the system's chaotic behavior and how it is affected by its initial conditions.

A double pendulum project in MATLAB is a great way to explore the fascinating world of nonlinear dynamics and chaos. By simulating the motion of a simple physical system, you can gain insights into some of the most complex and interesting phenomena in science and mathematics.

We will see some graphs. They are:

I. Omega vs time & Theta vs time;
II. Phase Space;
III. Theta1 vs Theta2;
IV. Energy vs time;
V. Momentum vs time.

Theory:



Input:

- L1 length of the first bob from hanging point;
- L2 length of the second bob from first bob;
- M1 and M2 representing the mass of each pendulum bob;
- Initial angle $\theta 1$ of the first bob;
- Initial angle $\theta 2$ of the second bob;
- Time duration t.

Output:

We will see some graphs as output. They are:

I. Omega vs time & Theta vs time;
II. Phase Space;
III. Theta1 vs Theta2;
IV. Energy vs time;
V. Momentum vs time.

Mathematical Formulation of the Problem:

To arrive at a function of the angle of both bobs with respect to time, the double pendulum system will first be analyzed through the sum of forces applied on each bob. First, we will formulate the equations displacement, velocity and acceleration (Neumann):

$$x_1 = L_1 sin(\theta_1)$$
$$y_1 = -L_1 cos(\theta_1)$$
$$x_2 = x_1 + L_2 sin(\theta_2)$$
$$y_2 = y_1 - L_2 cos(\theta_2)$$

Differentiating to get the x and y velocities:

$$\dot{x_1} = \dot{\theta_1} L_1 cos(\theta_1)$$
$$\dot{y_1} = \dot{\theta_1} L_1 sin(\theta_1)$$
$$\dot{x_2} = \dot{x_1} + \dot{\theta_2} L_1 cos(\theta_2)$$
$$\dot{y_2} = \dot{y_1} + \dot{\theta_2} L_1 sin(\theta_2)$$

Differentiating again to get the x and y accelerations:

$$\ddot{x_1} = -\dot{\theta}_1^2 L_1 sin(\theta_1) + \ddot{\theta}_1 L_1 cos(\theta_1)$$
$$\ddot{y_1} = \dot{\theta}_1^2 L_1 cos(\theta_1) + \ddot{\theta}_1 L_1 sin(\theta_1)$$
$$\ddot{x_2} = \ddot{x_1} - \dot{\theta}_2^2 L_2 sin(\theta_2) + \ddot{\theta}_2 L_2 cos(\theta_2)$$
$$\ddot{y_2} = \ddot{y_1} + \dot{\theta}_2^2 L_2 cos(\theta_2) + \ddot{\theta}_2 L_2 sin(\theta_2)$$

Assuming the two bobs are represented as point particles, we will then analyze the forces applied on each bob:

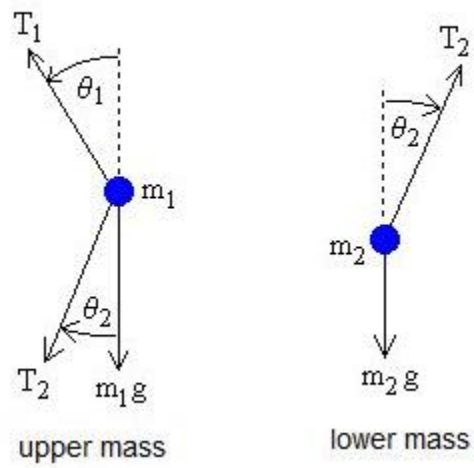


Figure 2: Free body diagrams of both the upper and lower bobs in the double pendulum system

Using Newton's second law of motion in the x and y directions on the first bob:

$$F_{x1} = m_1 a_{x1}$$
$$m_1 \ddot{x_1} = -T_1 sin(\theta_1) + T_2 sin(\theta_2)$$
$$F_{y1} = m_1 a_{y1}$$
$$m_1 \ddot{y_1} = T_1 cos(\theta_1) - T_2 cos(\theta_2) - m_1 g$$

Using Newton's second law of motion in the x and y directions on the second bob:

$$F_{x2} = m_2 a_{x2}$$
$$m_2 \ddot{x}_2 = -T_2 sin(\theta_2)$$
$$F_{y2} = m_2 a_{y2}$$
$$m_2 \ddot{y}_2 = T_2 cos(\theta_2) - m_2 g$$

By solving the previous equations for $\ddot{\theta}_1$ and $\ddot{\theta}_2$, we arrive at the following:

$$\ddot{\theta}_1 = \frac{-g(2m_1 + m_2)sin(\theta_1) - m_2 g sin(\theta_1 - 2\theta_2) - 2sin(\theta_1 - \theta_2)m_2(\dot{\theta}_2^2 L_2 + \dot{\theta}_1^2 L_1 cos(\theta_1 - \theta_2))}{L_1(2m_1 + m_2 - m_2 cos(2\theta_1 - 2\theta_2))}$$

$$\ddot{\theta}_1 = \frac{2sin(\theta_1 - \theta_2)(\dot{\theta}_1^2 L_1(m_1 + m_2) + g(m_1 + m_2)cos(\theta_1) + \dot{\theta}_1^2 L_2 m_2 cos(\theta_1 - \theta_2))}{L_2(2m_1 + m_2 - m_2 cos(2\theta_1 - 2\theta_2))}$$

$$\dot{\theta}_1 = \omega_1$$
$$\dot{\theta}_2 = \omega_2$$

We arrive at the following first order differential equations:

$$\dot{\omega}_1 = \frac{-g(2m_1 + m_2)\sin(\theta_1) - m_2 g sin(\theta_1 - 2\theta_2) - 2sin(\theta_1 - \theta_2)m_2(\omega_2^2 L_2 + \omega_1^2 L_1 cos(\theta_1 - \theta_2))}{L_1(2m_1 + m_2 - m_2 cos(2\theta_1 - 2\theta_2))}$$

$$\dot{\omega}_2 = \frac{2sin(\theta_1 - \theta_2)(\omega_1^2 L_1(m_1 + m_2) + g(m_1 + m_2)cos(\theta_1) + \omega_2^2 L_2 m_2 cos(\theta_1 - \theta_2))}{L_2(2m_1 + m_2 - m_2 cos(2\theta_1 - 2\theta_2))}$$

The equations are currently second order differential equations, but to be solved numerically the equation has to be in the form of a first order differential equation by applying the following substitution:

The double pendulum is a very interesting system as it is very simple but can show chaotic behavior for certain initial conditions. In this regime, slightly changing the initial values of the angles (θ_1,θ_2) and angular velocities ($\dot{\theta}_1$,$\dot{\theta}_2$) makes the trajectories of the bobs become very different from the original ones.

The Lagrangian for the double pendulum is given by L=K−U, where K and U are the kinetic and potential energies of the system respectively. The kinetic energy T is given by:

$$\begin{aligned} K &= \frac{1}{2}m_1 v_1^2 + \frac{1}{2}m_2 v_2^2 \\ &= \frac{1}{2}m_1(\dot{x}_1^2 + \dot{y}_1^2) + \frac{1}{2}m_2(\dot{x}_2^2 + \dot{y}_2^2) \\ &= \frac{1}{2}m_1 L_1^2\dot{\theta}_1^2 + \frac{1}{2}m_2\left[L_1^2\dot{\theta}_1^2 + L_2^2\dot{\theta}_2^2 + 2L_1L_2\dot{\theta}_1\dot{\theta}_2\cos(\theta_1 - \theta_2)\right] \end{aligned}$$

where above we used the fact that $\cos\ \theta_1 \cos\ \theta_2 + \sin\ \theta_1 \sin\ \theta_2 = \cos(\theta_1 - \theta_2)$. The potential energy U is given by:

$$\begin{aligned} U &= m_1 g y_1 + m_2 g y_2 \\ &= -m_1 g L_1 \cos\ \theta_1 - m_2 g(L_1 \cos\ \theta_1 + L_2 \cos\ \theta_2) \\ &= -(m_1 + m_2) g L_1 \cos\ \theta_1 - m_2 g L_2 \cos\ \theta_2 \end{aligned}$$

The Lagrangian of the system is then:

$$\begin{aligned} L = \frac{1}{2}(m_1 + m_2)L_1^2\dot{\theta}_1^2 \quad &+ \frac{1}{2}m_2 L_2^2\dot{\theta}_2^2 + m_2 L_1 L_2 \dot{\theta}_1\dot{\theta}_2\cos(\theta_1 - \theta_2) \\ &+ (m_1 + m_2) g L_1 \cos\ \theta_1 + m_2 g L_2 \cos\ \theta_2 \end{aligned}$$

The canonical momenta associated with the coordinates θ_1 and θ_2 can be obtained directly from L

$$\begin{aligned} p_{\theta_1} &= \frac{\partial L}{\partial \dot{\theta}_1} = (m_1 + m_2)L_1^2\dot{\theta}_1 + m_2 L_1 L_2 \dot{\theta}_2 \cos(\theta_1 - \theta_2) \\ p_{\theta_2} &= \frac{\partial L}{\partial \dot{\theta}_2} = m_2 L_2^2 \dot{\theta}_2 + m_2 L_1 L_2 \dot{\theta}_1 \cos(\theta_1 - \theta_2) \end{aligned}$$

Description of numerical techniques has been applied:

Euler's Method:

Euler's method is a simple numerical method used to approximate the solution of an ordinary differential equation (ODE) with a given initial value. The method is based on the idea of approximating the solution of the ODE by the solution of a first-order Taylor series approximation.

The method is iterative and consists of the following steps:

a. We have to define the ODE as y' = f(x,y), where y(x) is the unknown function and f(x,y) is a given function.
b. Then we have to choose a step size h and a starting point (x_0,y_0).
c. Then using the following iterative formula to obtain an approximation of the solution y at the
    a. next step (x0+h):
    b. $y_1 = y_0 + h*f(x_0, y_0)$
d. Repeat step "c" to obtain the solution at the desired points.

Euler's method is a first-order method, meaning that the error of the approximation is proportional to h. Thus, it is not very accurate for small values of h. However, it is a simple and easy-to-implement method, and it can serve as a starting point for more advanced numerical methods.

In our project we use the code:

```
alpha1 = @(theta1,theta2,omega1,omega2) (-g*(2*m1+m2)*sin(theta1) - m2*g*sin(theta1-2*theta2) - 2*sin(theta1-theta2)*m2*(omega2^2*L2+omega1^2*L1*cos(theta1-theta2))) / (L1*(2*m1+m2-m2*cos(2*theta1-2*theta2)));

 alpha2 = @(theta1,theta2,omega1,omega2)(2*sin(theta1-theta2)*(omega1^2*L1*(m1+m2)+g*(m1+m2)*cos(theta1)+omega2^2*L2*m2*cos(theta1-theta2))) / (L2*(2*m1+m2-m2*cos(2*theta1-2*theta2)));


% Solve using Euler's method
for i = 1:N-1

   omega1(i+1) = omega1(i) + alpha1(theta1(i),theta2(i),omega1(i),omega2(i))*dt;
   omega2(i+1) = omega2(i) + alpha2(theta1(i),theta2(i),omega1(i),omega2(i))*dt;
   theta1(i+1) = theta1(i) + omega1(i+1)*dt;
   theta2(i+1) = theta2(i) + omega2(i+1)*dt;


end
```

Improved Euler's Method:

The Improved Euler's method (also known as Heun's method) is a modification of Euler's method for numerically approximating the solution of an ordinary differential equation (ODE) with a given initial value. This method improves upon Euler's method by taking into account the curvature of the solution.

The method is iterative and consists of the following steps:

a. We have to define the ODE as y' = f(x,y), where y(x) is the unknown function and f(x,y) is a given function.
b. Then we have to choose a step size h and a starting point (x0,y0).
c. Then using Euler's method to obtain an intermediate approximation of the solution at the next step:
   y* = y0 + h*f(x0, y0)
d. Using the intermediate approximation to estimate the slope at the next step:
   slope = (f(x0, y0) + f(x0 + h, y*))/2
e. Using the estimated slope to obtain a better approximation of the solution at the next step:
   y1 = y0 + h*slope
f. Repeat steps "c" to "e" to obtain the solution at the desired points.

Improved Euler's method is a second-order method, meaning that the error of the approximation is proportional to h^2. This makes it more accurate than Euler's method for small values of h. However, it is still a relatively simple and easy-to-implement method, and it can serve as a good compromise between accuracy and computational cost for many applications.

In our project we use the code:

```
% Solve using Improved Euler's method
 for i = 1:N-1


omega1(i+1)=omega1(i)+(dt/2)*(alpha1(theta1(i),theta2(i),omega1(i),omega2(i))+alpha1(theta1(
i+1),theta2(i+1),(omega1(i)+alpha1(theta1(i),theta2(i),omega1(i),omega2(i)).*dt),omega2(i+1)));


omega2(i+1)=omega2(i)+(dt/2)*(alpha2(theta1(i),theta2(i),omega1(i),omega2(i))+alpha2(theta1(
i+1),theta2(i+1),omega1(i+1),(omega2(i)+alpha2(theta1(i),theta2(i),omega1(i),omega2(i)).*dt)));

   theta1(i+1) = theta1(i) + omega1(i+1)*dt;

   theta2(i+1) = theta2(i) + omega2(i+1)*dt;


end
```

Function of APP:

1. After opening the APP we will see this kind of interphase.

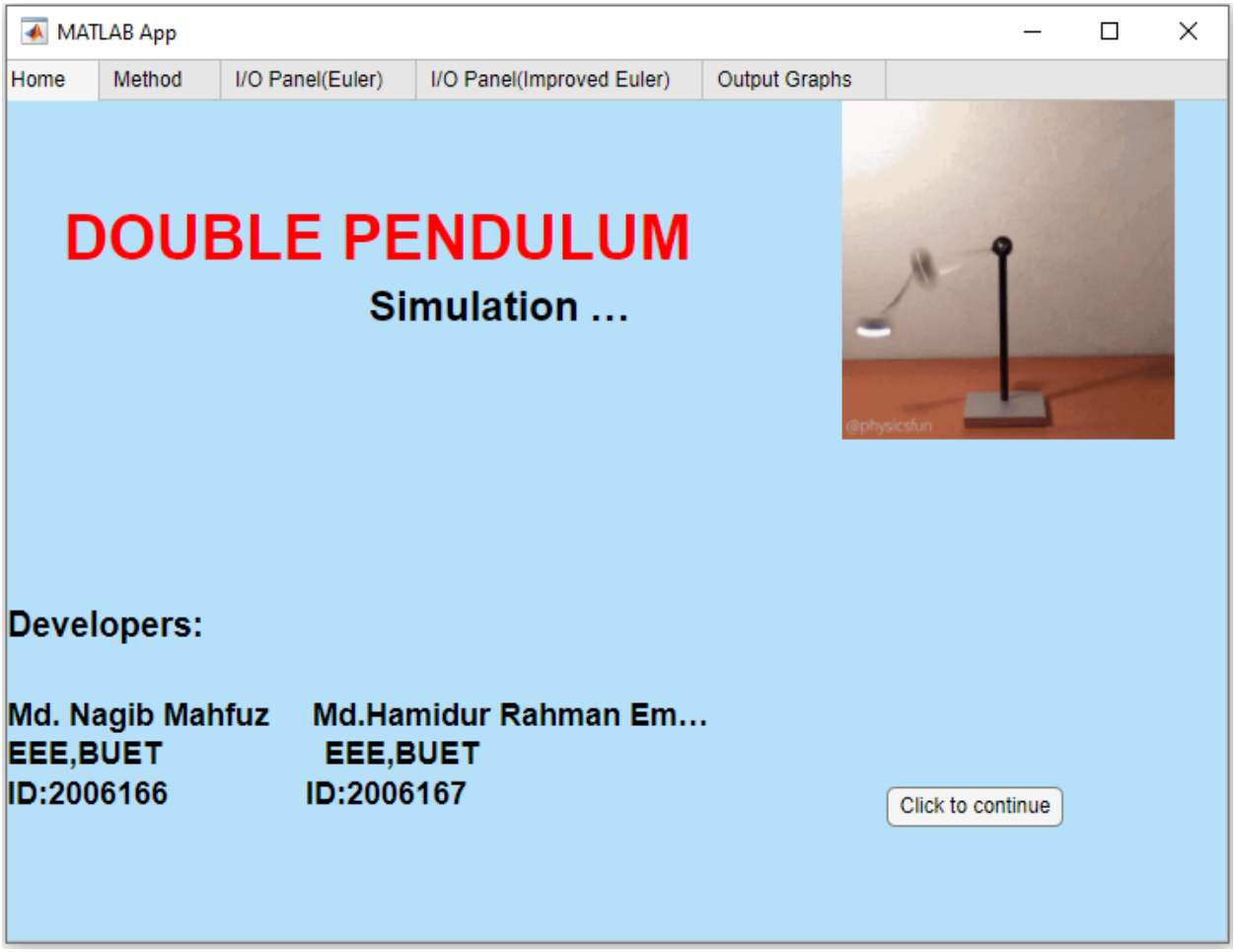


Figure: Home Page

To go to the next page we have to click the button “Click to continue”.

2. Then, a page will be shown containing Methods.

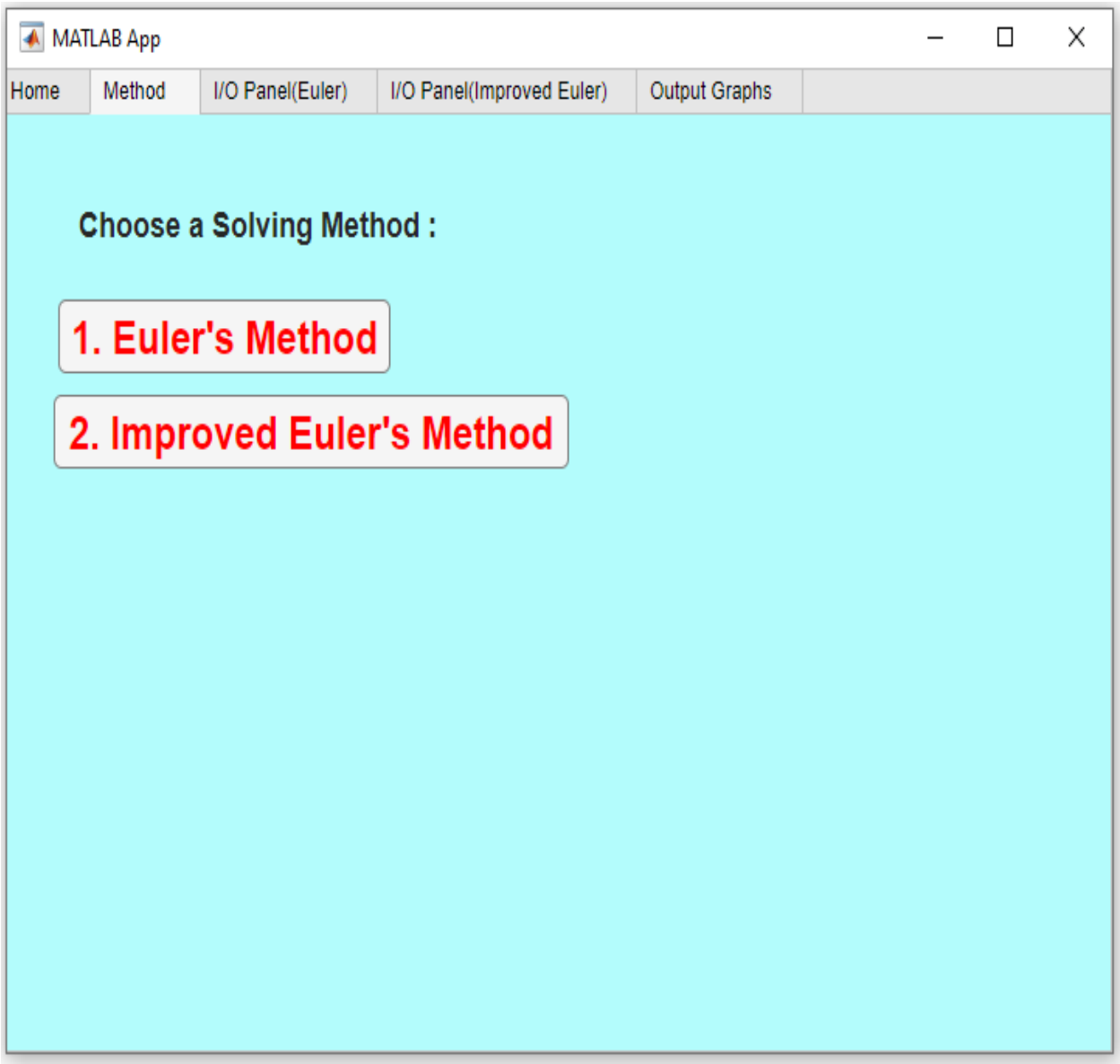


Figure: Method

3. We have to choose a method to go ahead. Let's select the option 1, Euler's Method. This interphase will be shown.

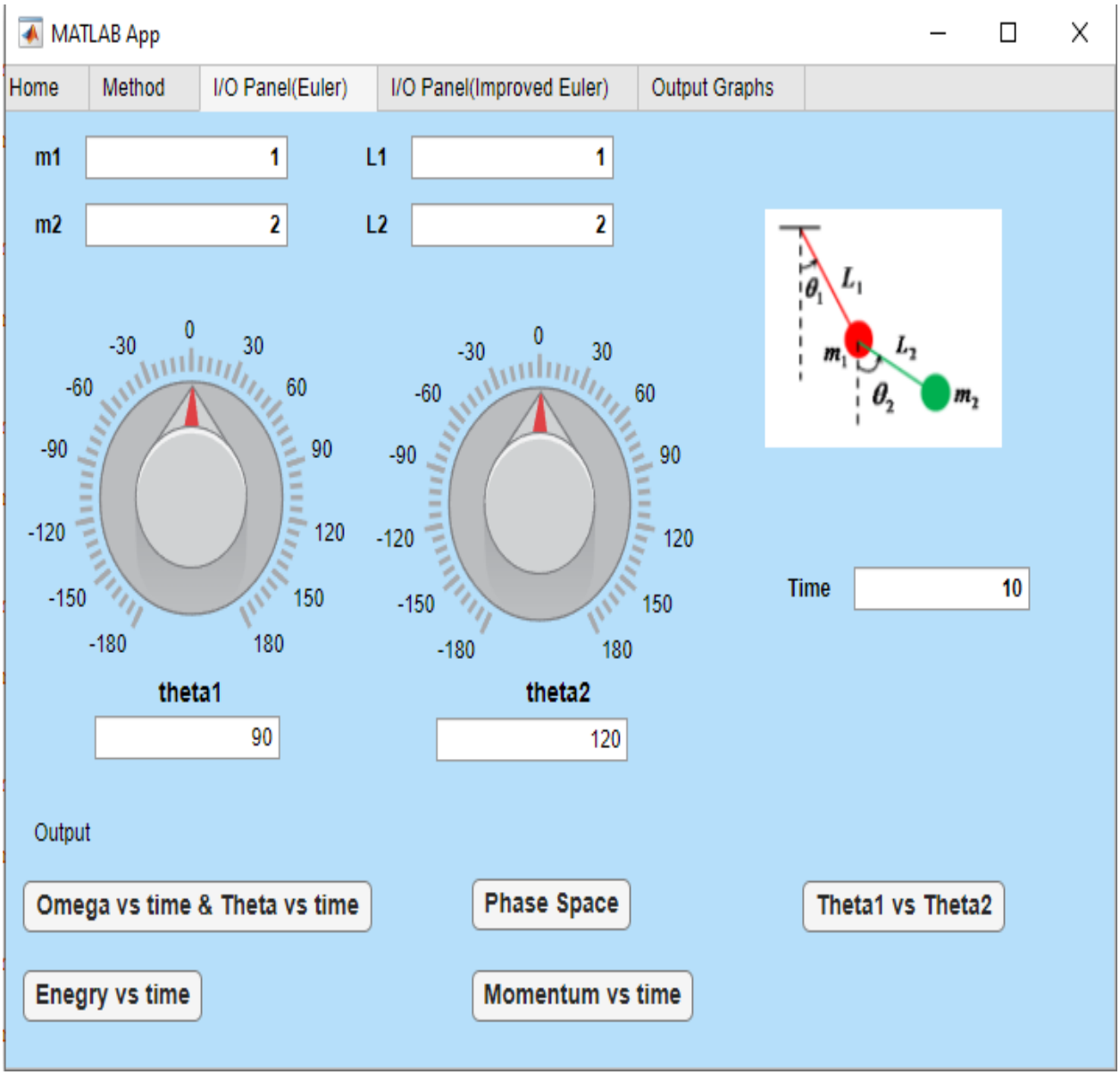


Figure: I/O Panel(Euler)

Let's put some test values in the respective data field.

We will do the same thing for Improve Euler's method. And also put the same data as the Euler's method.

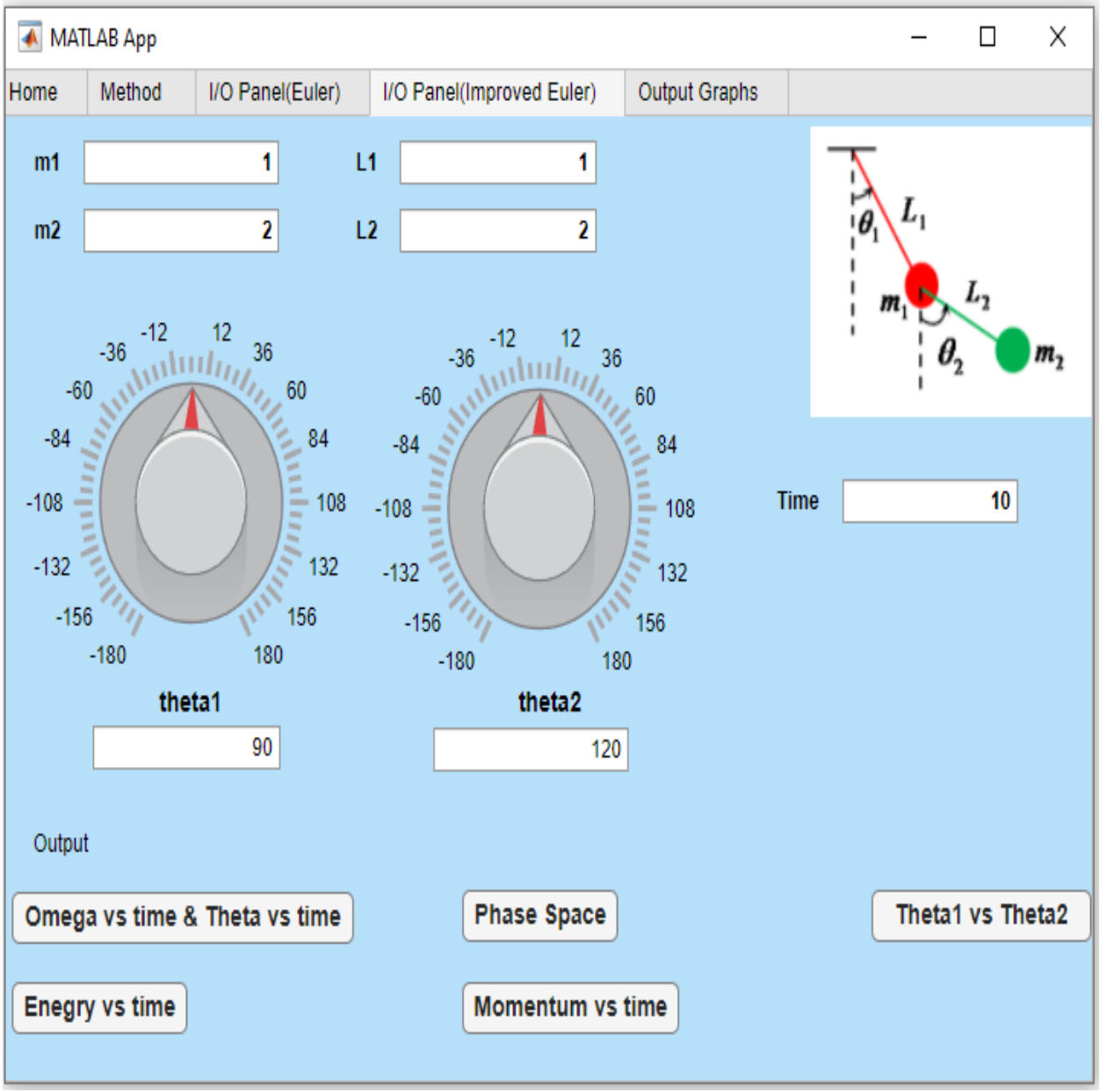
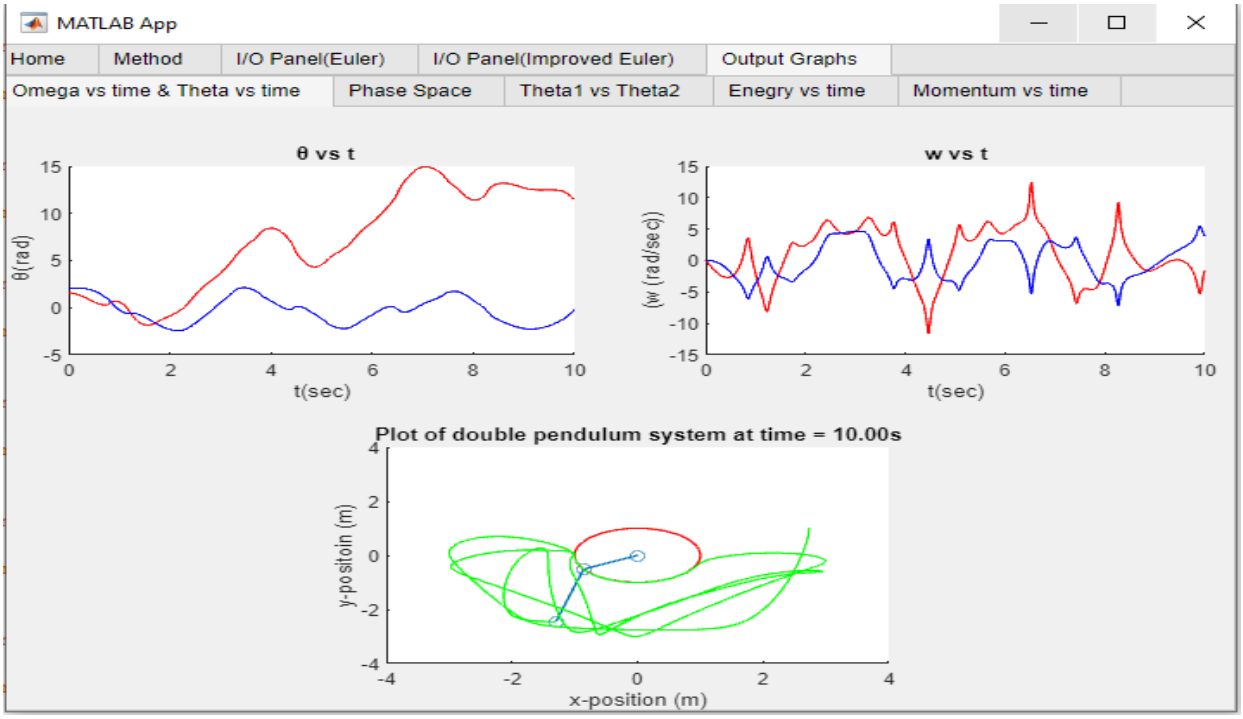
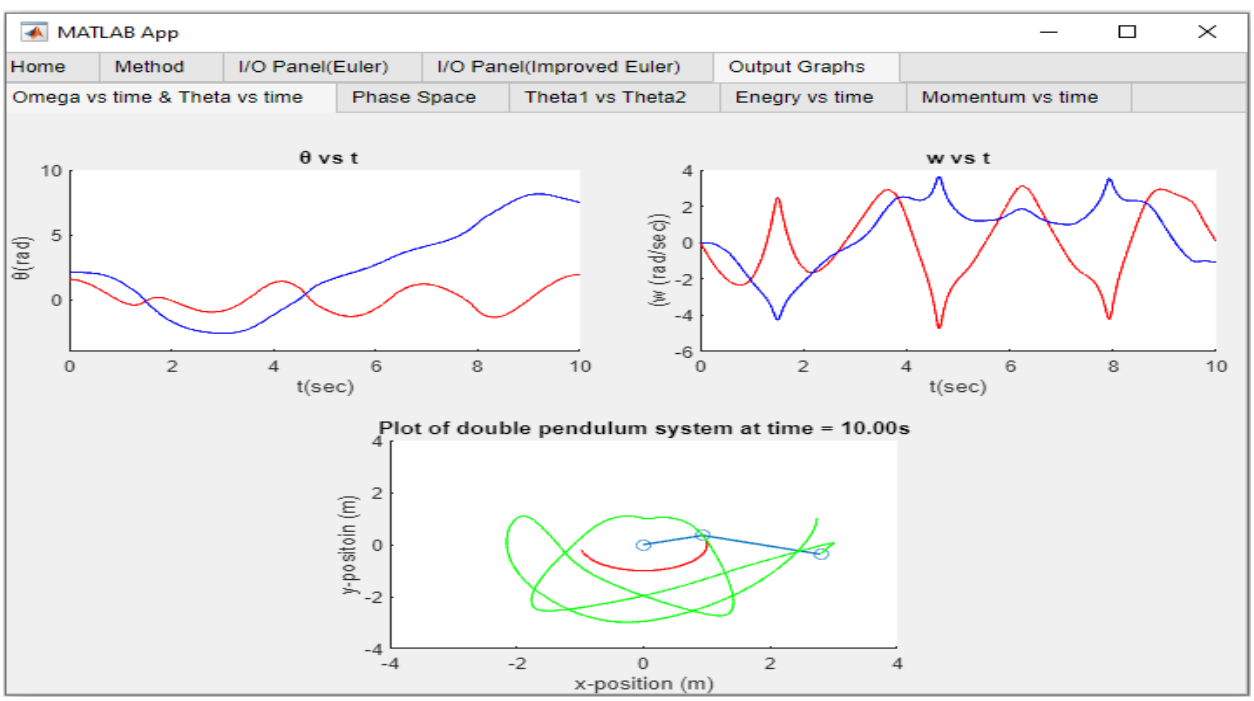


Figure: I/O Panel(Improved Euler)

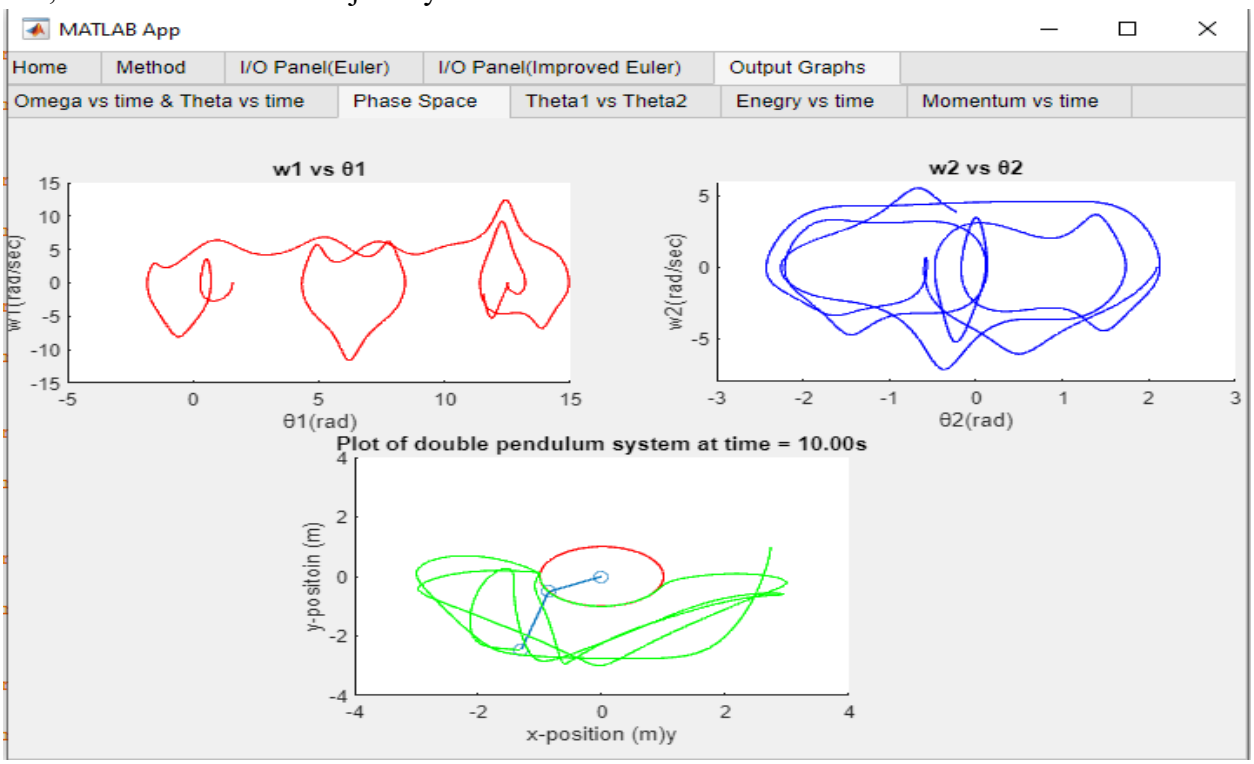
4. Then if we click the button “Omega vs time & Theta vs time” we will see the output curve containing “θ vs t” , “w vs t” and the trajectory .



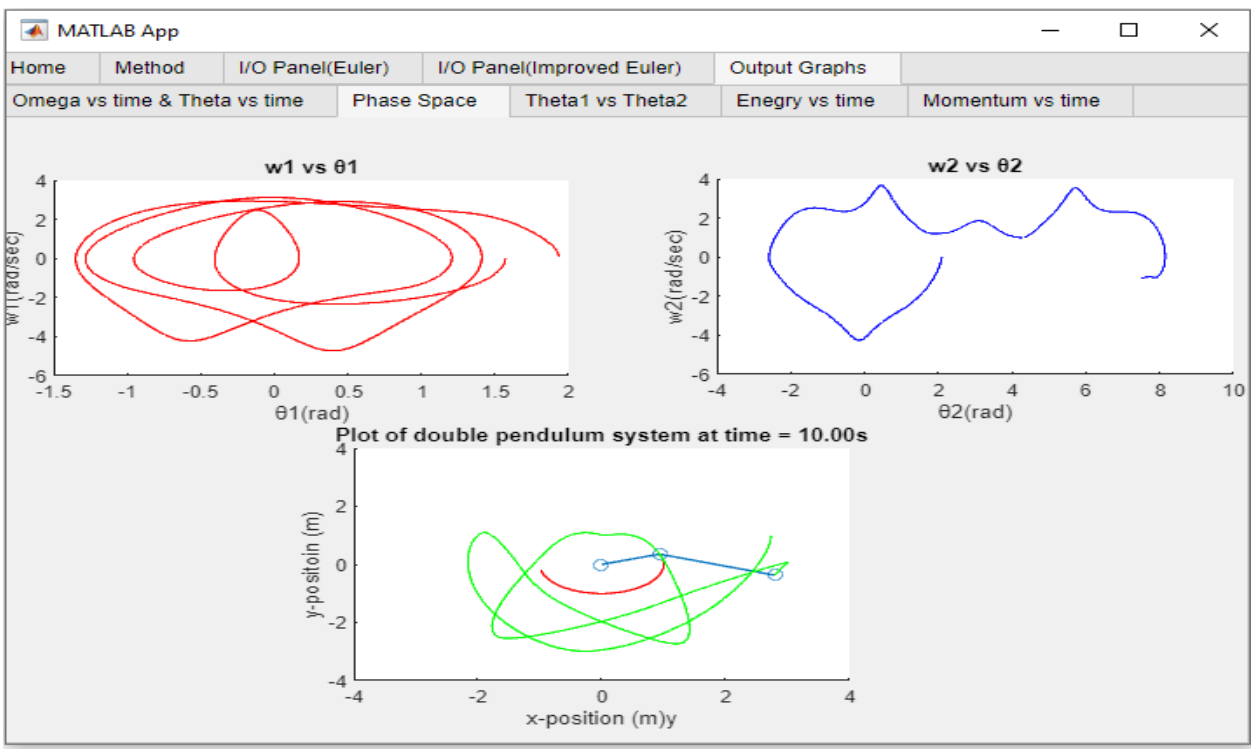
Similarly,we will get curve for Improved Euler’s method.



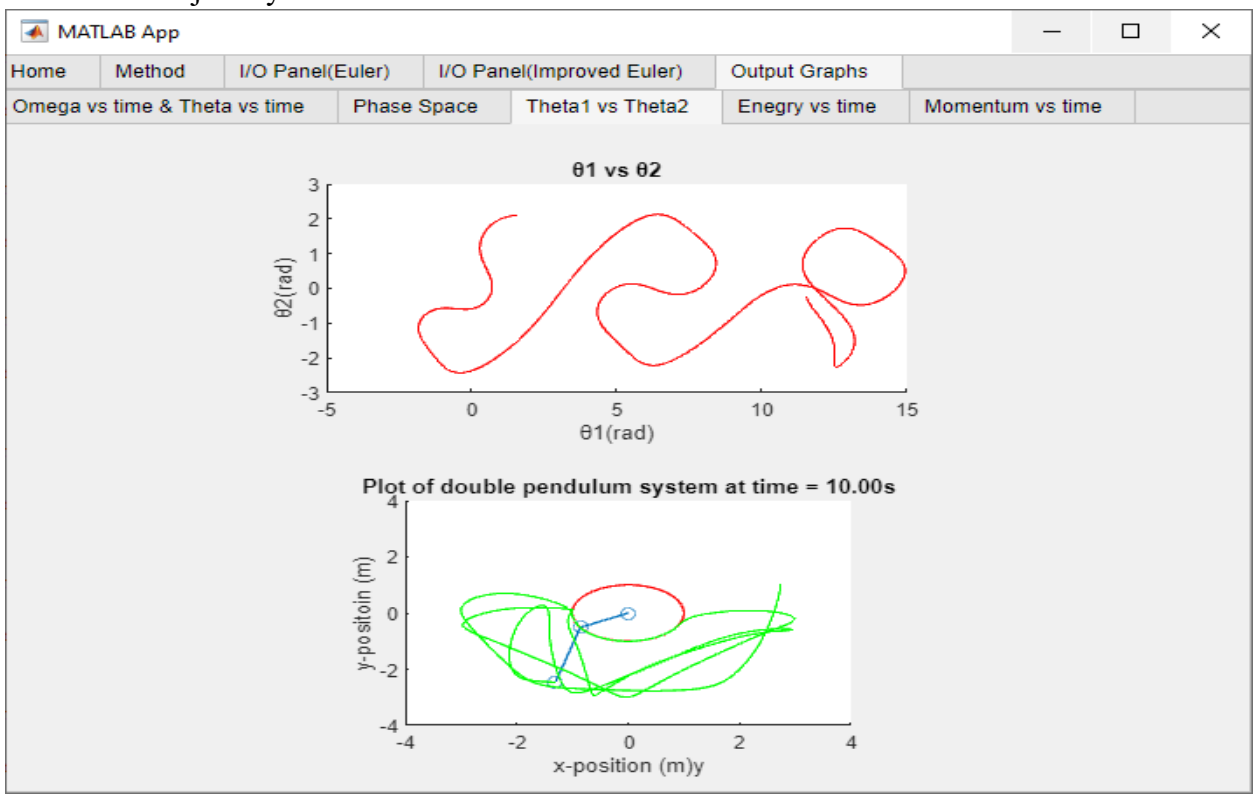
5. Then if we click the button “Phase Space” we will see the output curve containing “w1 vs θ1”, “w2 vs θ2” and the trajectory .



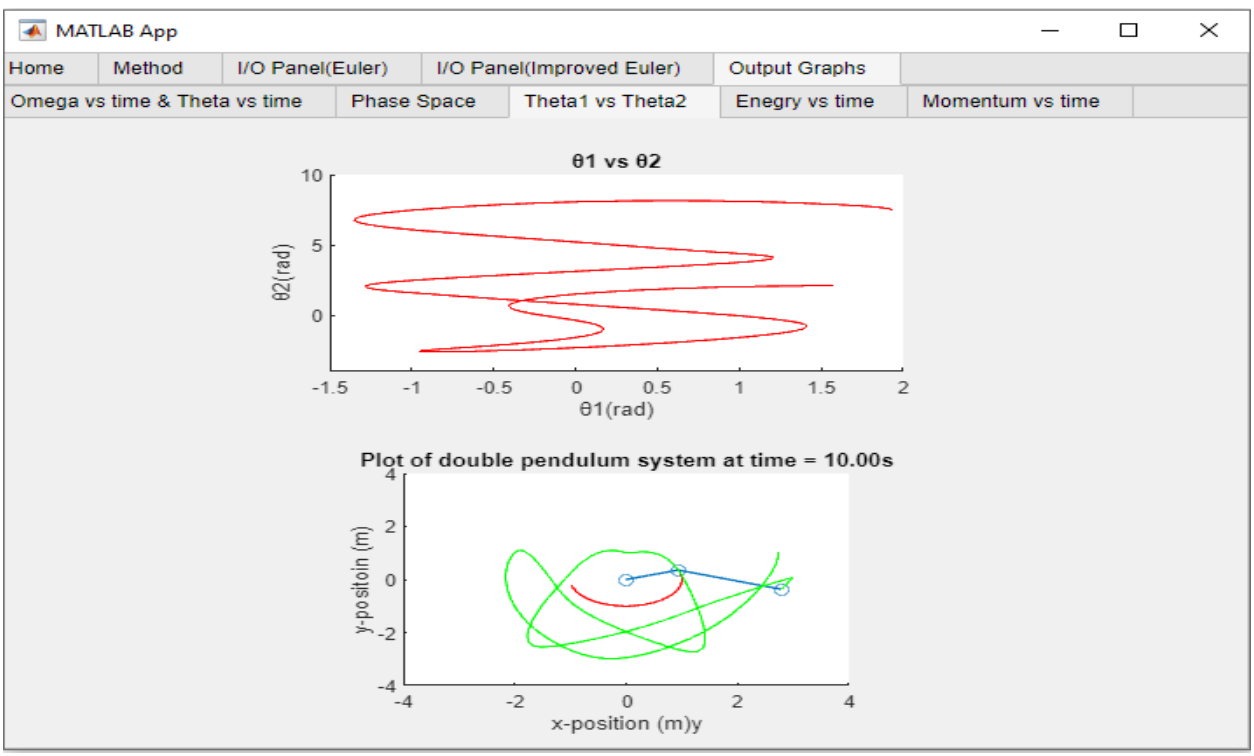
Similarly,we will get curve for Improved Euler’s method.



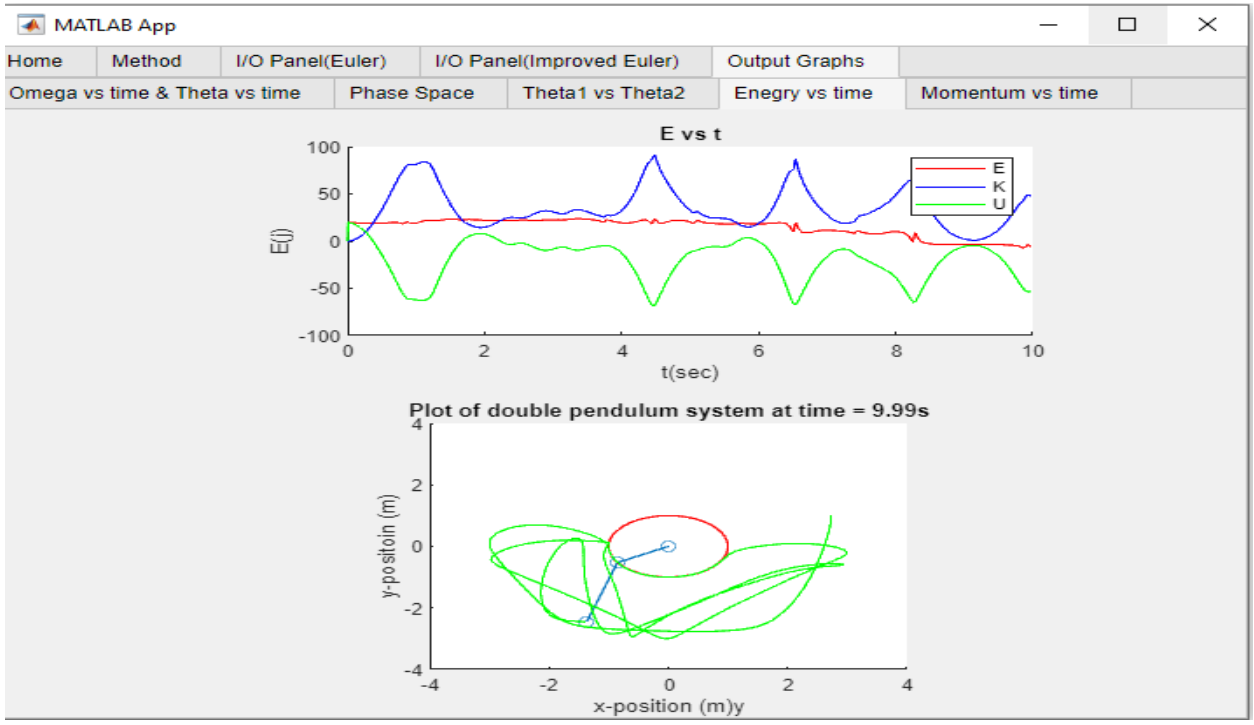
6. If we click the button “Theta1 vs Theta2” we will see the output curve containing “θ1 vs θ2” and the trajectory.



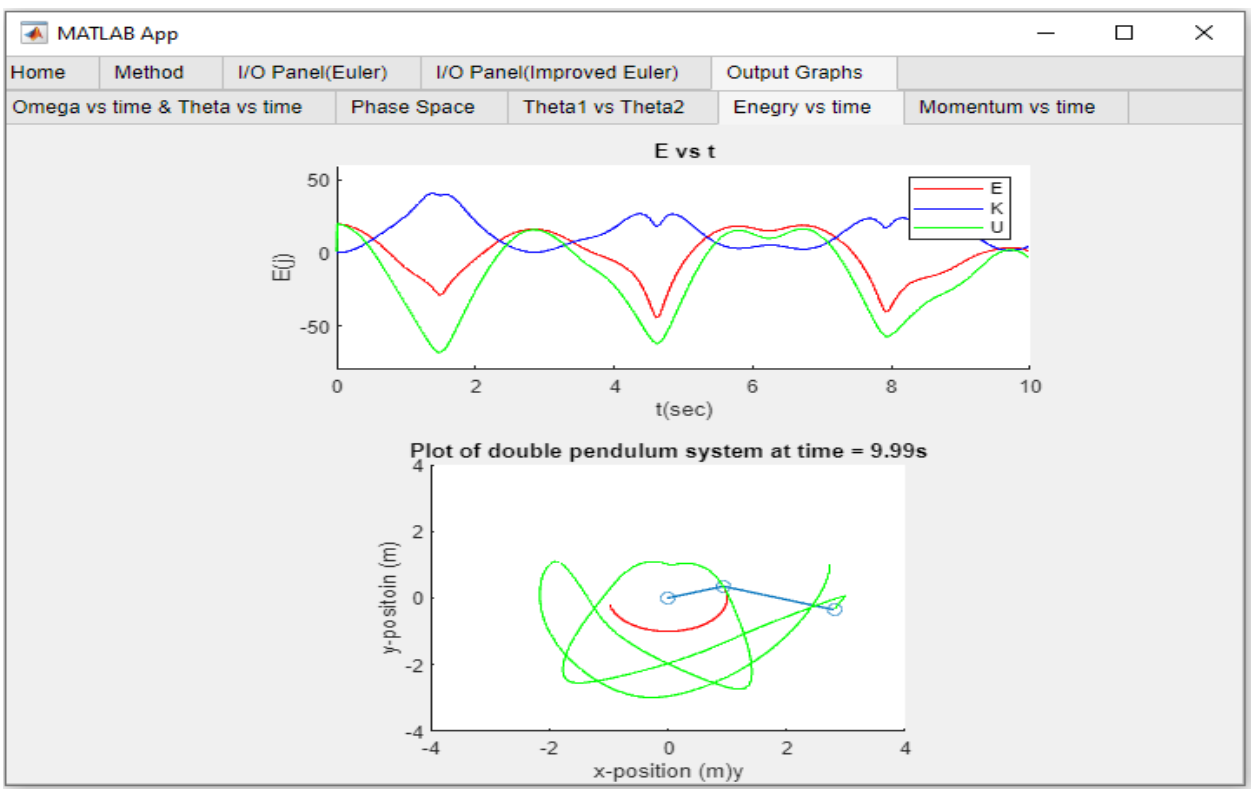
Similarly,we will get curve for Improved Euler’s method.



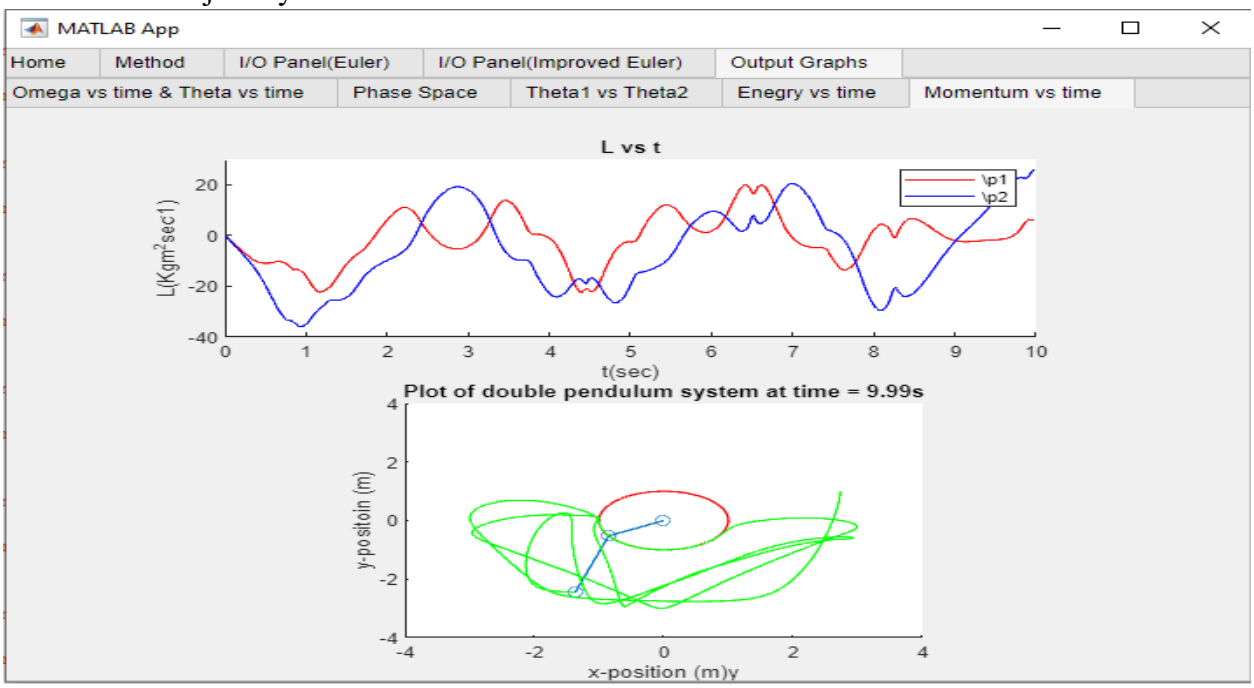
7. If we click the button “Energy vs time” we will see the output curve containing “E vs t” and the trajectory .



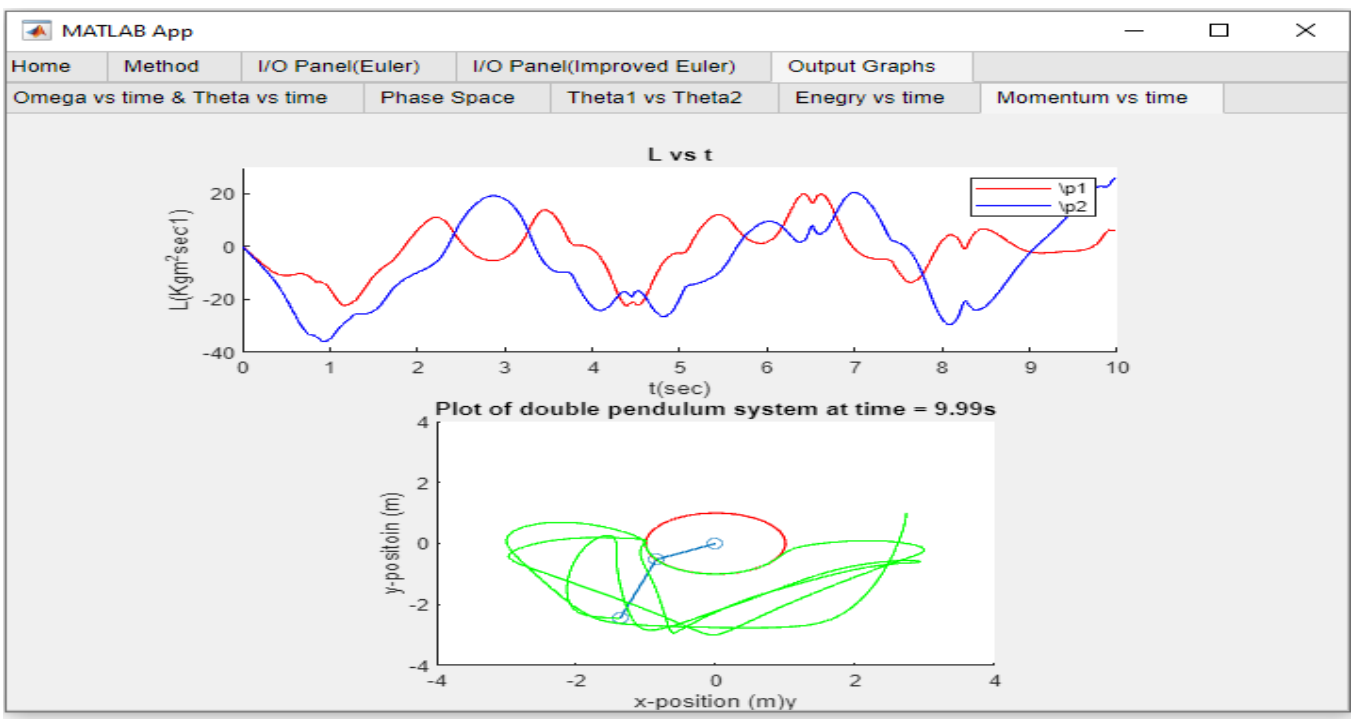
Similarly,we will get curve for Improved Euler’s method.



8. If we click the button “Momentum vs time” we will see the output curve containing “L vs t” and the trajectory .



Similarly,we will get curve for Improved Euler’s method.



Conclusion:

The double pendulum is a classic example of a chaotic system where small changes in the initial conditions can lead to significantly different outcomes over time. This was clearly demonstrated by the simulations using both the Euler and improved Euler methods, which showed that even a small difference in the initial angles of the pendulum can lead to drastically different behavior.

The Euler method is a simple but less accurate numerical integration technique that relies on a first-order Taylor series expansion. This method may not accurately capture the chaotic behavior of the double pendulum, leading to large errors in the results.

On the other hand, the improved Euler method, which is a second-order numerical integration technique, provides a more accurate solution by taking the average of two estimates. This method reduces the errors introduced by the Euler method and captures the chaotic behavior of the double pendulum more accurately.

This project underscores the importance of accurate numerical methods in the study of chaotic systems and highlights the sensitivity of these systems to small perturbations.

Project Link:
https://buetedu-my.sharepoint.com/:f:/g/personal/2006166_eee_buet_ac_bd/EnMFc_nfNphCspzz7GoDDVkB0caZy7_J3Q10CsyGWeU_wA?e=UUZy9K